

The Utility Problem

As the grid continues to evolve through the energy transition, data will be relied upon more and more by utility companies, namely their operations and planning teams. With this in mind, both the access to data and the accuracy of the data are paramount to running the grid reliably and optimally. With the evolution of the grid, more and more Internet of Things (IoT) sensors exist in distribution service areas. With potentially millions of IoT sensors managed by a single utility, issues with their state (health) and data communication are guaranteed and could be caused by a range of influences such as manufacturing quality, environmental conditions, and age of the component.

IoT devices like smart meters, line sensors, smart reclosers/breakers or smart inverters are distinct asset groups within the utility. These asset groups should have their own systematic approach to their state monitoring and health overview. If this aspect is neglected it might influence multiple areas of the IoT operation ranging from increased expenditures for their operation or data communication issues.

In the area of state monitoring and health overview it is important to follow indicators of correct operation of IoT devices. IoT health data like temperature, battery voltage, communication signal strength, moisture or other parameters are vital sources of information to determine if an IoT device is operating properly. IoT devices which are not in good state - poor IoT device health - can lead to multiple issues such as incomplete data or inaccurate data reading.

Incomplete or inaccurate data readings due to IoT health could result in numerous detrimental impacts. Due to missing or incorrect data, the operations team could make decisions that could cause the grid to run inefficiently or even more concerning, could overload the circuit causing component failure and/or outages. In addition, the future of the grid is incredibly complex and requires comprehensive planning. This planning is greatly dependent on accurate, historical data, which is often gathered from IoT devices. With the inability to access this data or if the data is inaccurate, the planning of the grid will be jeopardized, which could lead to enormous economic costs as well as potentially dangerous construction and operation.

Overall, IoT devices are able to provide utility companies with important data needed for the reliable operation of today's grid and the grid of the future. To achieve maximum value from the installed IoT device fleet, it is necessary to have a proper IoT health monitoring system in place to gain an overview of IoT health. With this system in place, the utility is able to operate the IoT devices in a more efficient manner.

The Data Solutions and Methods

Awesense's Energy Transition Platform offers simplified ways to analyze IoT health throughout the distribution area. Awesense Platform's unique linkage of geospatial data (GIS) with time-series data enables this type of analysis. With data populated in the Awesense Platform, the use case can be accessed in Awesense's True Grid Intelligence (TGI) solution (TGI Sensor

Alerts and TGI Grid Alerts) or the use case can be built using Energy Data Model's (EDM) interfaces.

The basic version of this use case is available in TGI. TGI implementation of this use case is simple but includes all necessary components to meet requirements. This can be done through two different mechanisms. The user can navigate to different sensors through the 'Sensors' page, click on the sensor name and analyze IoT health data like battery voltage level, signal strength of the sensor and missed samples in the last hour. Alternatively, for a more proactive approach, TGI comes with built-in alerts, such as:

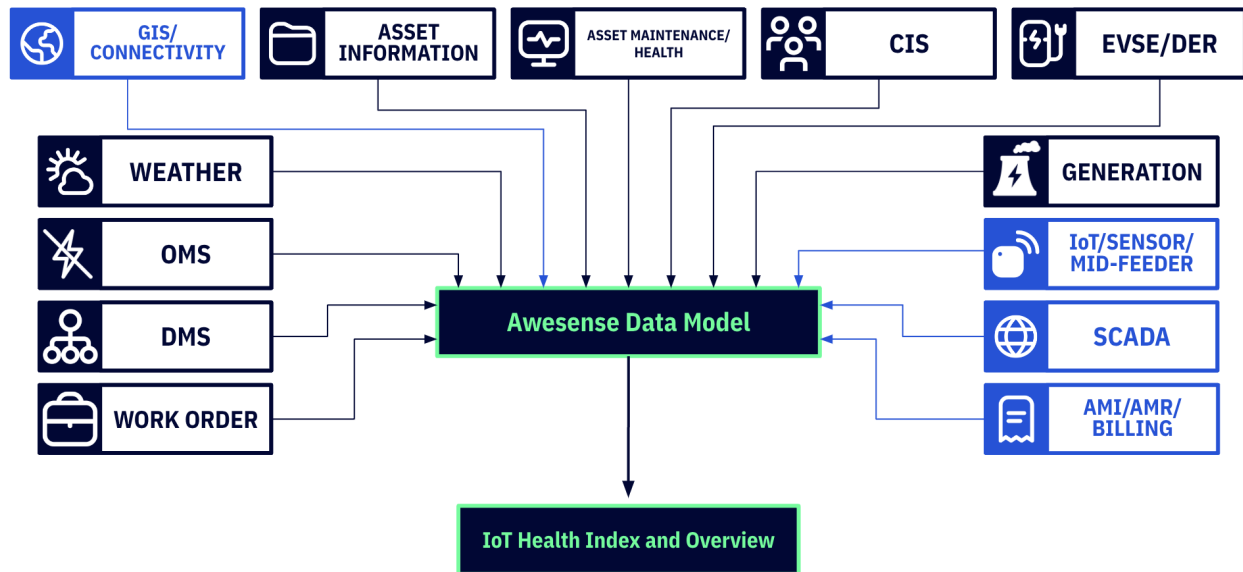
- "Missed Samples" - Sensor missed measurements in checked period.
- "Inconsistent Count" - Sensor sent a different number of packets than expected.
- "Communication Alert" - Sensor did not communicate with the server when scheduled to.

With these alerts set for the desired sensors, TGI will alert the user when a sensor is not operating as desired. This alert will include all of the sensor specifications and location so that the issue can be addressed.

If the analyst would like to build this use case from scratch, the analyst should first identify all of the IoT sensors on the grid. This can be achieved by filtering the GIS table for IoT sensors. Depending on how the analyst would like to segment the grid, the predefined Awesense functions such as *grid_get_connected()* and *grid_get_downstream()* can be utilized in conjunction with the element *type* in order to determine all desired sensors. These sensors can then be analyzed based on the *ts_data_source_double* table, which provides the timestamps for the recording and when it was received along with the period and value of the data points. All IoT sensor time-series are readily available in EDM data model. These data points can be analyzed together in order to determine the IoT health or determine the accurate readings consistent with the entire grid as well as proper communication of the sensor. Any inconsistencies will arise in the analysis, which can be provided to a key decision maker in order to improve the grid.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. TGI Sensor Alerts - Please speak to Awesense about access to the sandbox environment.
2. A GitHub link to the Jupyter notebook can be provided upon request.